

Scope of Work

Project Title: NFL Multi-Platform Trading Analysis Dashboard

Project Name: Market Making Analysis [MMA]

Date: 08-10-2025

1. Client's Requirement

The client requires a **unified trading dashboard** for professional traders who wish to trade NFL sports markets across multiple platforms. The system should:

- Fetch and display **live NFL market data from Pinnacle** (reference feed).
- Allow traders to **analyse odds** and **post corresponding quotes** on multiple platforms (e.g., **Kalshi, ProphetX**) via API integrations.
- Support **manual** as well as **automated trading**, where automation adjusts quotes and trade behaviour based on Pinnacle's live odds.
- Implement **risk management** controls (stop-loss, quote limits, exposure tracking).
- Record all trades and generate **real-time profit and loss (P&L)** summaries per user, market, and platform.
- Maintain **counterparty tracking** to monitor performance of traders (sharp vs weak bettors) for future quote adjustments.
- Provide **notifications and logs** for user alerts, trade confirmations, and system events.

Essentially, the platform acts as a **smart bridge** between Pinnacle (reference market) and other trading platforms, automating quote posting and ensuring safe, risk-controlled operations.

2. Developer's Solution and Deliverables

We will build an end-to-end system that fulfils the above goals with the following deliverables:

A. Core Features

1. User Authentication & Access

- Secure login via username/password.
- API key registration for automated Quoting and Quote Posting.

2. Live Market Feed Integration (Pinnacle)

- Establish WebSocket & REST connection to Pinnacle's live NFL data feed.
- Cache live odds and market information for rapid internal access.

3. Target Platform Connectivity

- Integrate APIs for Kalshi and ProphetX and other mentioned Platforms.
- Enable quote posting, modification, and cancellation directly from the dashboard.

4. Trading & Automation Engine

- Allow manual quote posting or enable auto-trading based on Pinnacle's odds.
- Apply user-defined automation rules (spread difference, liquidity conditions, etc.).
- Log every trade for audit and future reference.

5. Risk Management Layer

- Enforce user-level and market-level limits (e.g., quote size, position exposure, stop-loss).
- Dynamic restriction based on trader or counterparty behaviour.

6. Profit & Loss Tracking

- Real-time P&L per market, per user, per platform.
- Historical performance analytics.

7. Counterparty Tracking

- Maintain historical behaviour of bettors interacting with trader quotes.
- Identify "sharp" or "weak" bettors to optimize quote size and spread width in future trades.

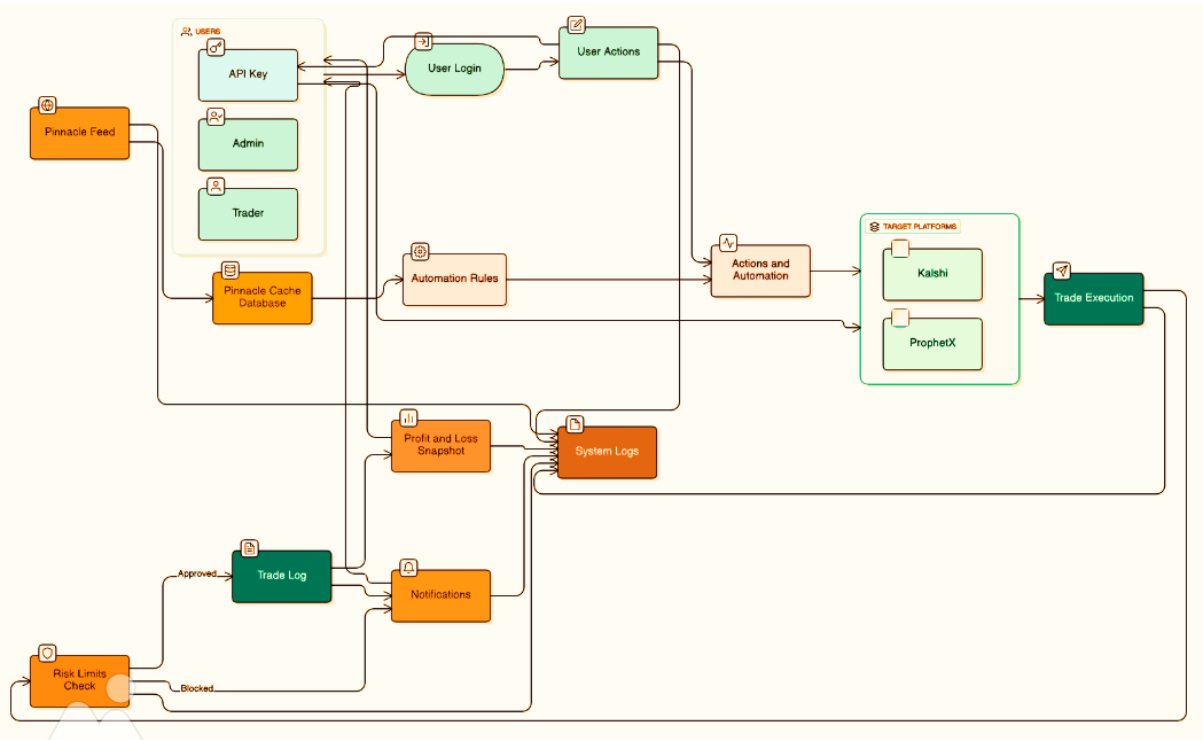
8. Monitoring, Notifications & System Logs

- Real-time alerts for trade executions, errors, and system anomalies.
- Central log system for debugging and audit trail.

9. Web Dashboard Interface

- Intuitive frontend showing markets, odds, automation controls, trades, and alerts.
 - Separate views for traders and admins.
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3. Data Flow Charts



4. Proposed Tech Stack

Layer	Technology / Tools	Purpose
Frontend	React.js	User dashboard and trading UI
Backend	Python (Fast API)	Core API server handling trading logic
Database	PostgreSQL	Structured storage for users, trades, limits, logs
Caching / Queue	Redis	Pinnacle feed caching and background job handling
Automation / Workers	Celery / Async task	Handle scheduled quote posting and updates
APIs Integration	REST & WebSocket	Connect with Pinnacle, Kalshi, ProphetX
Authentication	JWT / OAuth	Secure login and API token management
Deployment	Docker + Nginx	Containerized, scalable deployment

Layer	Technology / Tools	Purpose
Monitoring / Logging	Prometheus, Grafana	System health and trade activity visualization

5. Project Milestones

	Milestone	Description	Deliverables / Subtasks
1	Project Initialization & Environment Setup	Set up local and cloud environments for development.	- Create Git repo and folder structure - Setup virtual environment (venv / poetry) - Install FastAPI, PostgreSQL, SQLAlchemy, Alembic, Uvicorn - Initialize. env for secrets and configs
2	Database Schema Design & Creation	Design normalized PostgreSQL schema for markets, quotes, trades, risk, users, and platforms.	- Create DB ER diagram - Define tables for users, platforms, markets, quotes, orders, positions, risk limits, logs, notifications, etc. - Run migrations via Alembic
3	Backend Foundation (FastAPI Base)	Create backend structure for modular API development.	- Create main.py, api/, models/, schemas/, services/, utils/, routes/ - Setup dependency injection & response models - Test /health route
4	Pinnacle Integration (Market Data Feed)	Integrate Pinnacle live odds feed and store parsed data in DB.	- Create Pinnacle API client with WebSocket or REST - Parse and normalize odds to canonical market schema - Display on /pinnacle/live endpoint
5	Kalshi API Exploration & Authentication Testing	Study Kalshi API docs, test authentication, and access private APIs.	- Test login, markets, orderbook, and trade endpoints - Verify rate limits, pagination, and response normalization.
6	UMS (User Management System) & Authentication	Build secure user system and JWT authentication.	- Create /auth/register, /auth/login, /auth/refresh routes - Integrate PostgreSQL user storage - Implement roles (admin, trader) - Add middleware for route protection

7	Minimal Frontend Prototype (Vercel Demo)	Build minimal frontend dashboard (React + Tailwind) hosted on Vercel.	- Create 5 pages: Pinnacle Feed, Quotes, Risk Management, P&L, Logs/Notifications - Setup navigation layout and dummy data
8	Kalshi Integration for Manual Quoting	Enable posting quotes manually on Kalshi from dashboard.	- Integrate quote posting API - Add validation for quote size, price, and market mapping - Store quote logs in DB
9	Risk Management Module (Backend)	Add backend logic for quote size, exposure, and stop-loss control.	- Define exposure calculation logic - Store per-trader and global risk limits - Real-time triggers on overexposure
10	Automation Engine (Auto Mover + Quote Adjuster)	Create automation logic that updates quotes dynamically based on bettor performance & exposure.	- Auto-mover engine that adjusts odds or size - Logging and simulation mode - Option to enable/disable auto mode per market
11	Profit & Loss (P&L) Visualization	Add real-time P&L tracking and visualization dashboard.	- Track open & closed positions - Calculate realized/unrealized P&L - Use Chart.js / Recharts for visualization
12	Counterparty Analytics Dashboard	Build analytics dashboard showing counterparties' behaviour and patterns.	- Track sharp vs weak bettors - Aggregate statistics per counterparty - Display charts (win rate, quote volume, ROI trends)
13	Logs & Notification System	Build structured log and alerting module for system activity.	- Store logs for trades, quotes, and API events - Add notification service (DB + WebSocket) - Add frontend notifications panel
14	Deployment (Primary Phase - Kalshi)	Deploy backend + frontend integration for production testing.	- Deploy FastAPI on VM (Docker + Nginx + SSL) - Connect via domain mma.onestopanalytics.com
15	System Optimization &	Improve performance and scalability for real-time trading.	- Add caching (Redis) - Implement async DB operations

	Advanced Enhancements		- Introduce background tasks (Celery / FastAPI Tasks) - Add Prometheus + Grafana monitoring
16	Add Multi-Platform Expansion (ProphetX / Polymarket)	Extend system to integrate additional platforms easily.	- Abstract common APIs into adapter pattern - Plug in new platform modules seamlessly
17	Comprehensive Testing & QA	End-to-end testing and optimization.	- Unit + Integration testing (pytest) - Load & latency testing - Final security audit
18	Final Deployment & Maintenance	Final deployment, monitoring setup, and documentation.	- Docker Compose for full stack - Monitoring dashboard - Developer & trader documentation

6. Expected Outcome

The final system will allow traders to:

- Seamlessly compare live Pinnacle NFL odds and post optimized quotes to Kalshi and ProphetX.
- Enable automation rules that trade hands-free while respecting defined risk limits.
- Gain analytical insight through P&L, counterparty stats, and historical logs.
- Operate in a secure, scalable, and fully auditable environment.